

CS 111

Foundations of Program Design
Week 5, Wednesday Lecture

Quiz 4

Review for midterm exam

Material covered so far:

- Compiling Java code
- Branching programs
- Looping programs

Material covered on the exam:

- Everything above
- Reasoning about state
- Programs with multiple methods

Compiling Java code

The working directory

`javac / java` usage. Saving and modifying source code.

Printline debugging

`main()` entry point. `System.out.println()` to console. *Reading comments.*

Variable assignment

`int`, `boolean`, and `String` variables. Literal values.

Expressions

Combining literals with variables. Identifying the operation and its return type.

Branching programs

Using expressions

String concatenation. Integer division `/`, remainder operator `%`. Comparison operators, `>` `>=` `=` `<=` `<`.

Conditional statements

`if` / `else if` / `else`. Variable scope. *Boolean operators*, and `&&`, or `||`, not `!`.

Console input

Indexing into `String[] args`. Converting `String` to `int` using `Integer.parseInt()`.

Looping programs

Conditional loops

`while` and `for`. Counting values from `Math.random()`. Accumulating output over multiple loops with `String` concatenation.

Array assignments

`int[]` and `String[]` literals. Calculating the size of the array in order to initialize it.

Reasoning about state

Loops with multiple variables

Iterators, like `int i = 0; while (i < 5) { i += 1; }.`

Accumulators, like `String result = ""; result += someLongCalculation();`

Programs with multiple methods

Converting code into methods

Returning outputs from methods. Reading (but not modifying) arguments to a method.

Exception handling

`try` / `catch` syntax. Manipulating text as a `char[]` array.